

EBU6408 Sensors and Radio Frequency Identification (RFID)

Lecture 2: RFID Components

Part 3: Software Components

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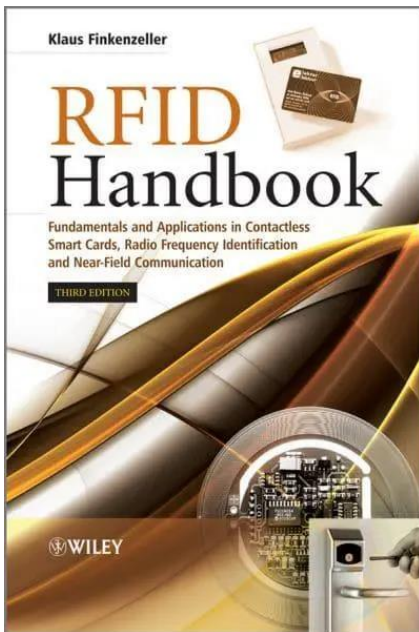
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Textbooks

Klaus Finkenzeller, “RFID Handbook”, Wiley, Third Edition, 2010.



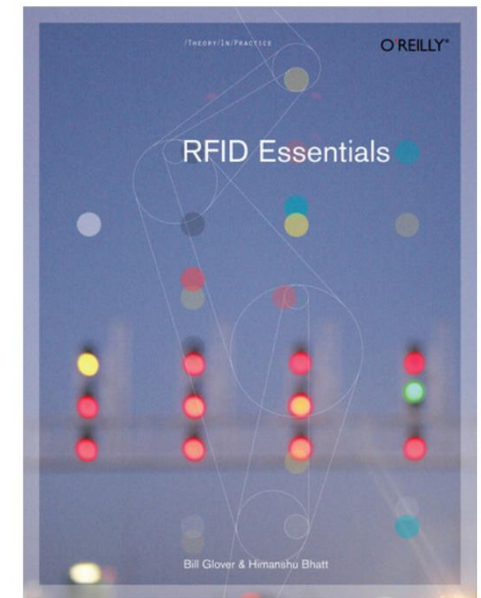
[PDF File](#)

Harvey Lehpamer, “RFID Design Principles”, Artech House, 2012.



[QMUL Library link](#)

Bill Glover and Himanshu Bhatt, “RFID Essentials”, O'Reilly Media Inc, 2006.



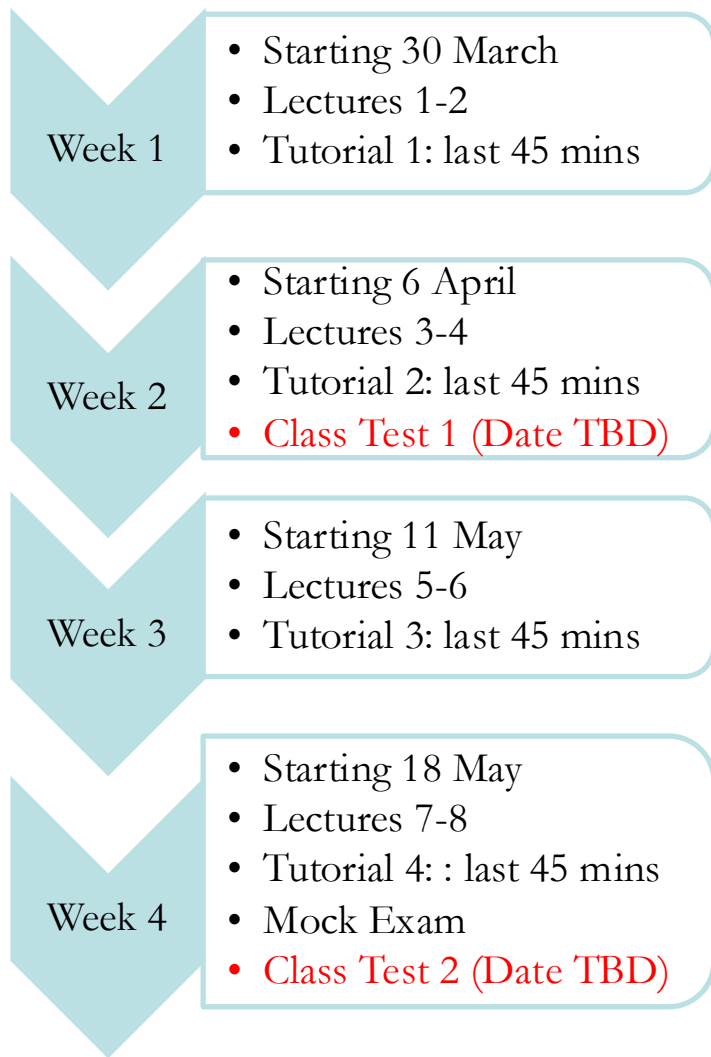
[QMUL Library link](#)

Evaluation Rules???

- **Two class tests:**
 - **Test 1:** Covering the material delivered in the first half of module **(5%)**
 - **Test 2:** covering the material delivered in the second half of module **(10%)**
 - **Allowed: : Handwriting double-faced A4-sized cheating sheet**
- **Coursework:** Work in groups of 4/5 students.
 - **Lab 1:** Introduction to Arduino **(Bonus Mark)**
 - ✓ Submission of written report on hardware design and programming code.
 - **Lab 2:** RFID projects using Arduino kits **(5%)**
 - ✓ Submission of written report on hardware design and programming code and **5-min video demonstration**
 - **Mini-project:** using practical knowledge from labs and chatbot support **(10%)**
 - ✓ Submission of written report on hardware design and programming code and **10-min video demonstration**
- **Final exam:** covering all teaching weeks **(70%)**
 - **Allowed: Handwriting double-faced A4-sized cheating sheet**
- A couple of QM+ online quizzes on teaching lectures **(Bonus Mark)**



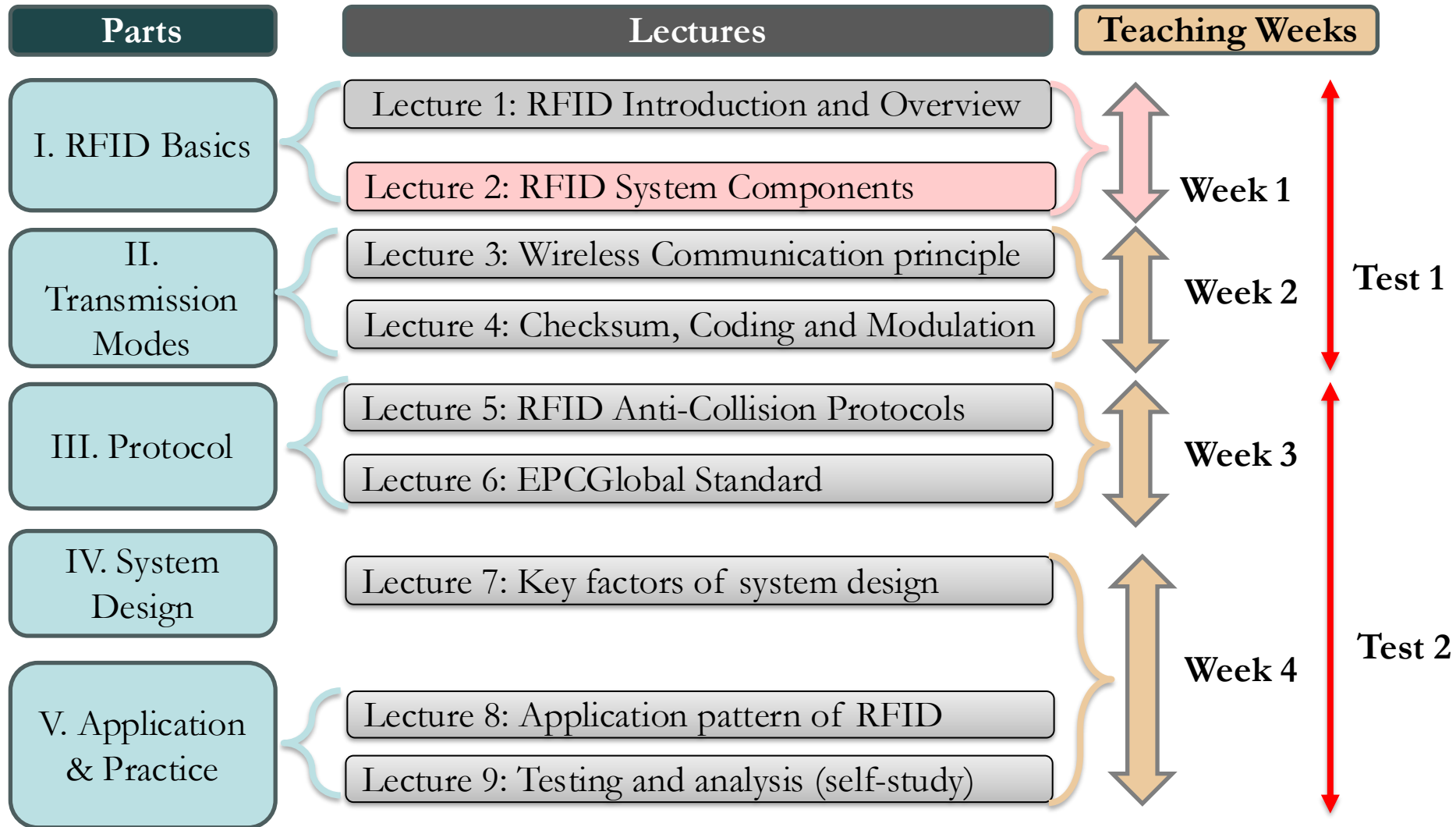
Module Activities



- Labs (Arduino):
 - Two lab sessions will take place in between the teaching weeks.
 - Work in groups (4/5 students) to share the hardware
 - Assignment submission deadlines: usually after lab session.
 - Mini-project due at the end of the term.
- Exact dates will be confirmed later on.

QM+ Quizzes will take place during term time for bonus marks

Module Content



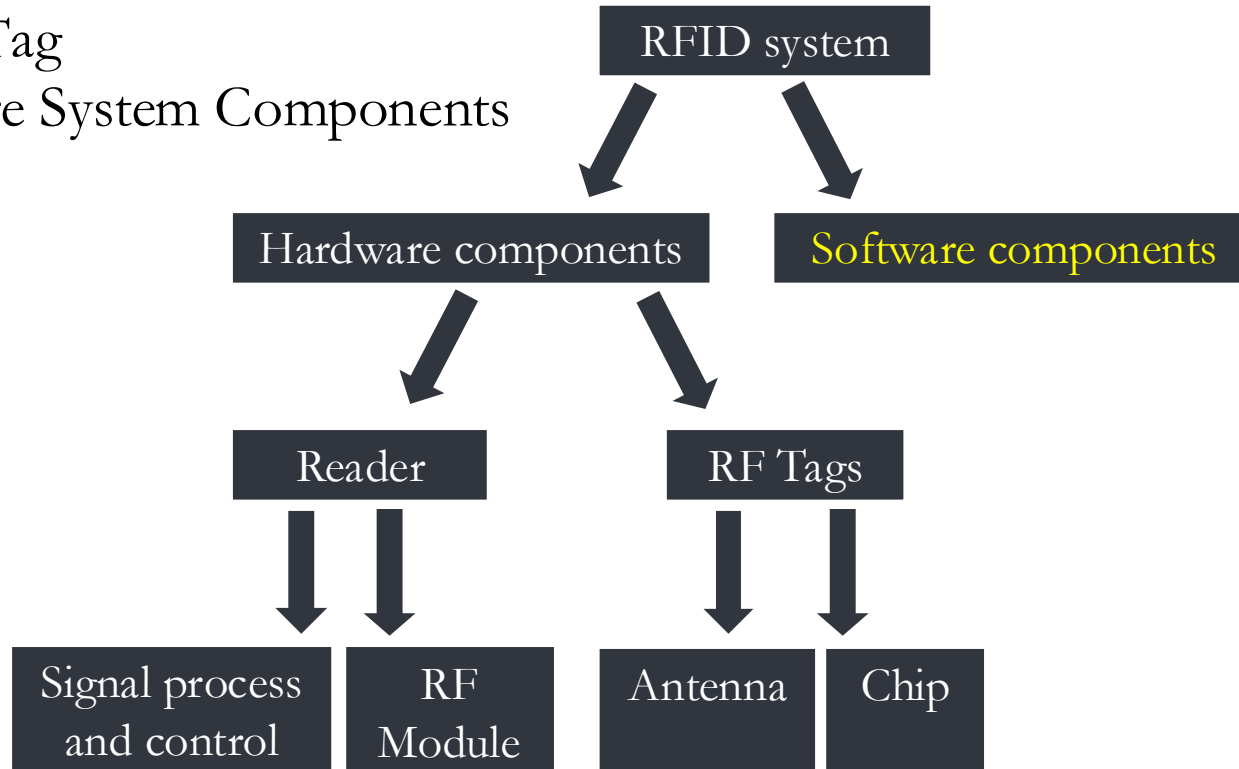
Recap

In last session, we learnt about:

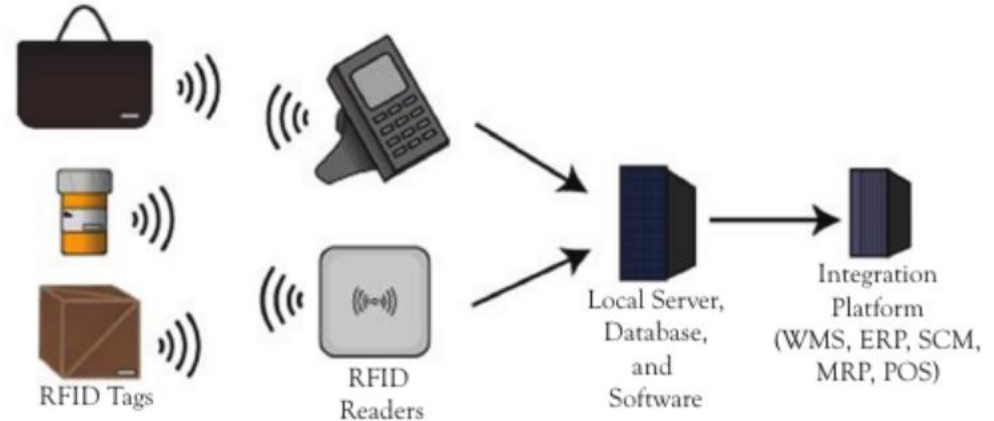
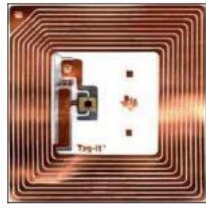
- ❖ What is an RFID
- ❖ Types of automatic identification technology
- ❖ Main features of RFID
- ❖ Core technology of RFID
- ❖ Brief History of RFID
- ❖ Development trends of RFID
- ❖ RFID and Internet of Things

Outline

- ❖ Reader
- ❖ RFID Tag
- ❖ Software System Components



Operating Principle of RFID Systems



- RFID typically consists of
 - A computer
 - A database
 - RFID software
 - A reader, which receives signals from the RFID and identifies and processes the information contained in one or more tags
 - RFID middleware
 - an antenna, used to transmit signals between the reader and the RFID tag.
 - An RFID tag (transponders), which contains the data of the element to identify
- The different for analysed designations of the tags and readers may vary, depending on applications, contexts, and their hardware and software resources.
- The readers are themselves connected to servers for data processing at middleware and application levels is, archiving and traceability

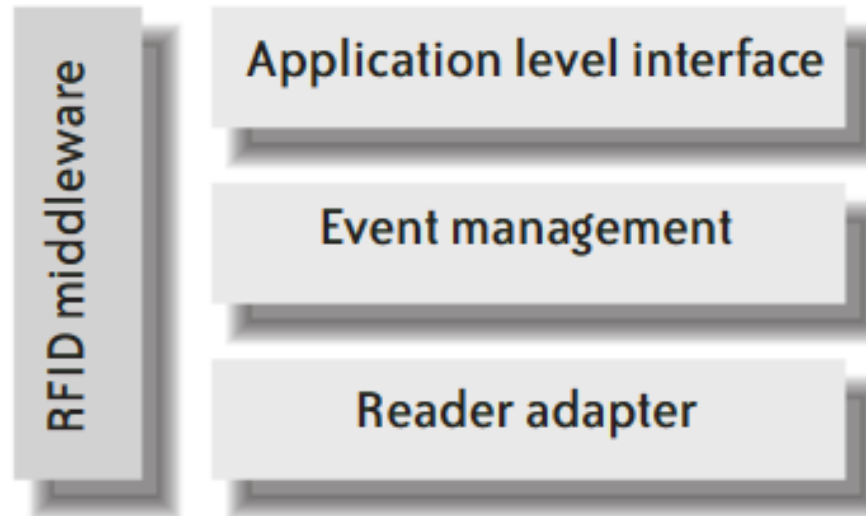
Section

RFID MIDDLEWARE



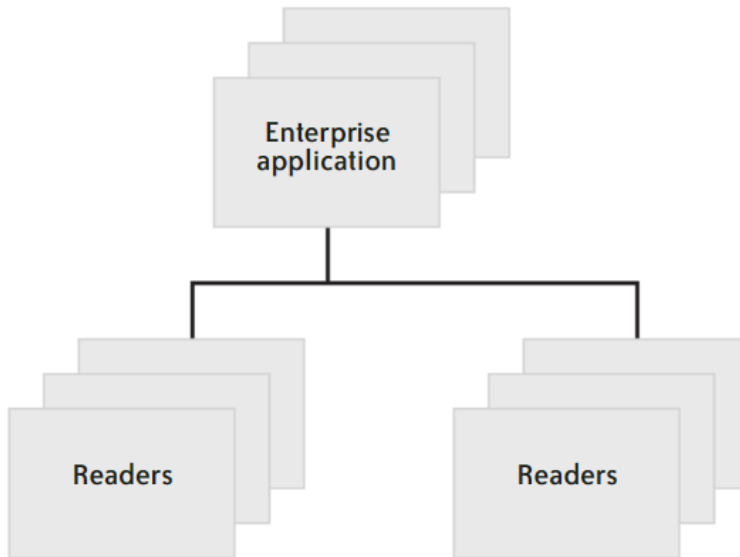
RFID Middleware

- ◆ One of the primary benefits of using RFID middleware is that it standardizes ways of dealing with the flood of information these tiny tags produce.

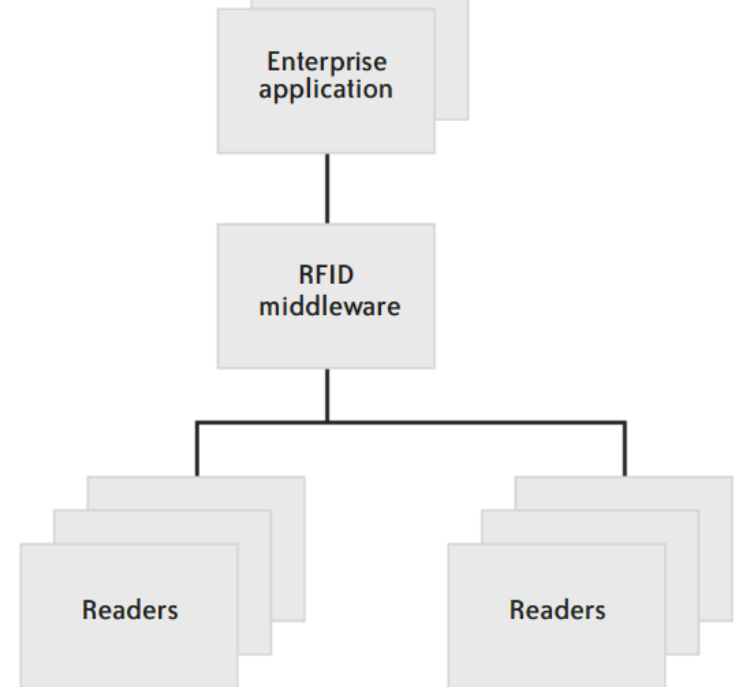


Motivations for Using RFID Middleware

- Three primary motivations for using RFID middleware:
 1. Providing connectivity with readers (via the **reader adapter**),
 2. Processing raw RFID observations for consumption by applications (via the **event manager**),
 3. Providing an **application-level interface** to manage readers and capture filtered RFID events.



Exposing applications directly to RFID readers



Using RFID middleware

SOFTWARE SYSTEM COMPONENT (SELF-STUDY)



Software System Component



The software components are the soul of RFID system.

To make readers have a direct experience of developing RFID, the commercial reader software interface will be introduced.

Overview

Workflow

Alien RFID Java
middleware

Reader use
examples

Overview



This part will introduce the software components of RFID system, especially middleware components.

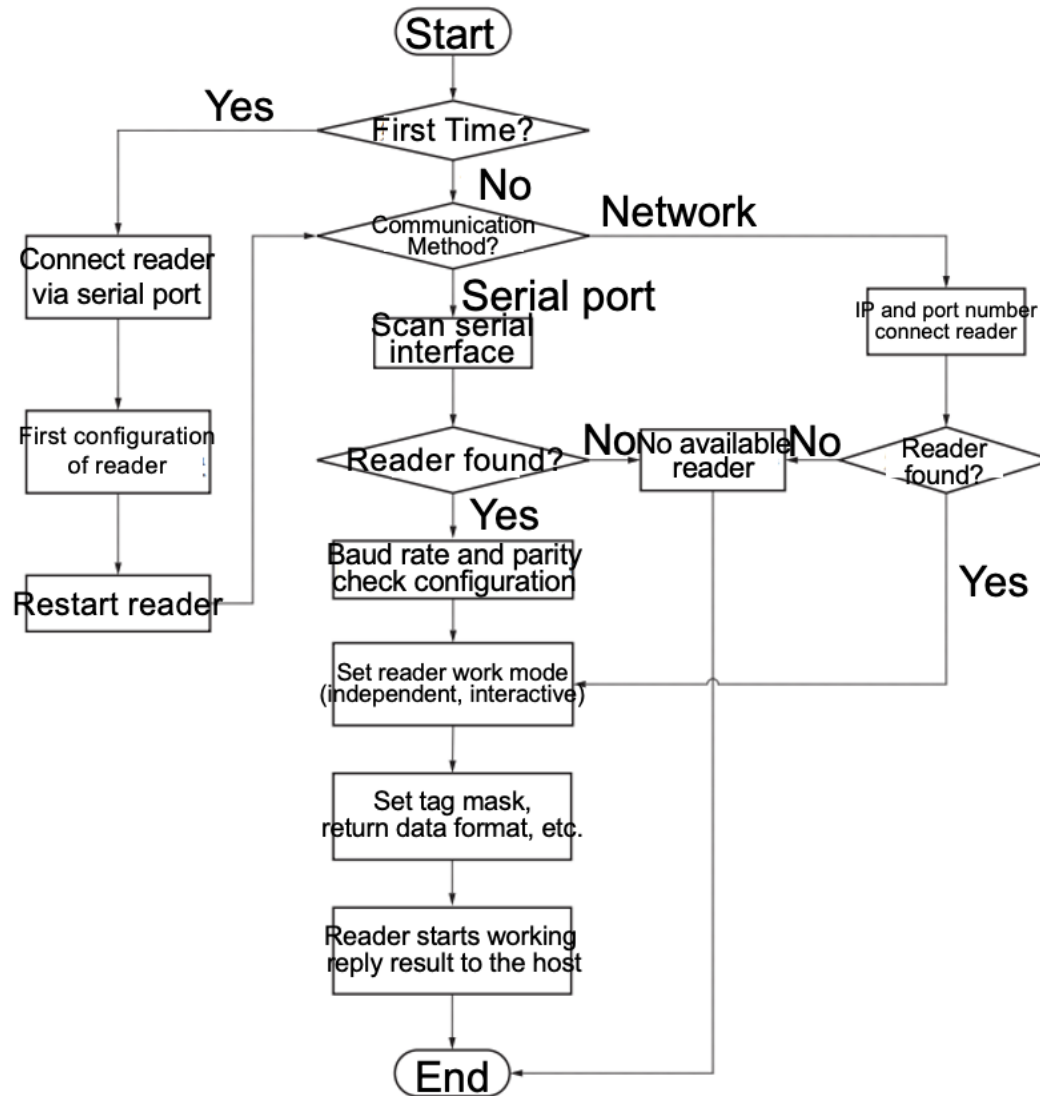
Take **UHF commercial reader** of **Alien Technology** for example.

Content:



The reader's initialization configuration of the installation process, tag mask set, the realization of independent work, the realization of interactive work, the storage structure and the wake-up methods.

Work Process



Alien RFID Java middleware

Organizational structure

Include 5 package: reader, tags, discovery, notify, util

Reader's connection

AlienClass1Reader class create object to control, with COM or network:

```
AlienClass1Reader reader = new AlienClass1Reader();  
reader.setConnection("COM1");  
AlienClass1Reader reader = new AlienClass1Reader();  
reader.setConnection("114.212.85.173", 23);  
reader.setUsername("alien");  
reader.setPassword("password");
```



Alien RFID Java middleware

Open or close connection

```
reader.open();  
reader.isOpen();  
reader.close();
```

Send operational command to reader

Through reader object's doReaderCommand

```
String readerName = reader.doReaderCommand("get  
ReaderName");  
  
// 获得阅读器读取范围内的标签列表  
public Tag getTagID(String tagID); Tag [] tagList = reader.getTagList();  
// 获得单个标签的详细信息  
public String getTagInfo(String tagID);  
// 读取标签用户存储区的数据  
public short[] getMemory(String tagID, int lengthIndex,  
int startIndex);  
// 向标签用户数据区内写入数据  
public boolean setMemory(String tagID, int startIndex,  
byte byteArray[]);  
// 设置标签掩码  
void setMask(String maskString);
```

Alien RFID Java middleware

Tag info class

Through getTagList function of reader object

```
Tag[] tagList = reader.getTagList();
```

```
public String getTagID();//return tag ID
```

```
public int getRSSI();//return signal strength
```

Search available reader

Serial service listener class:

SerialDiscoveryListenerService

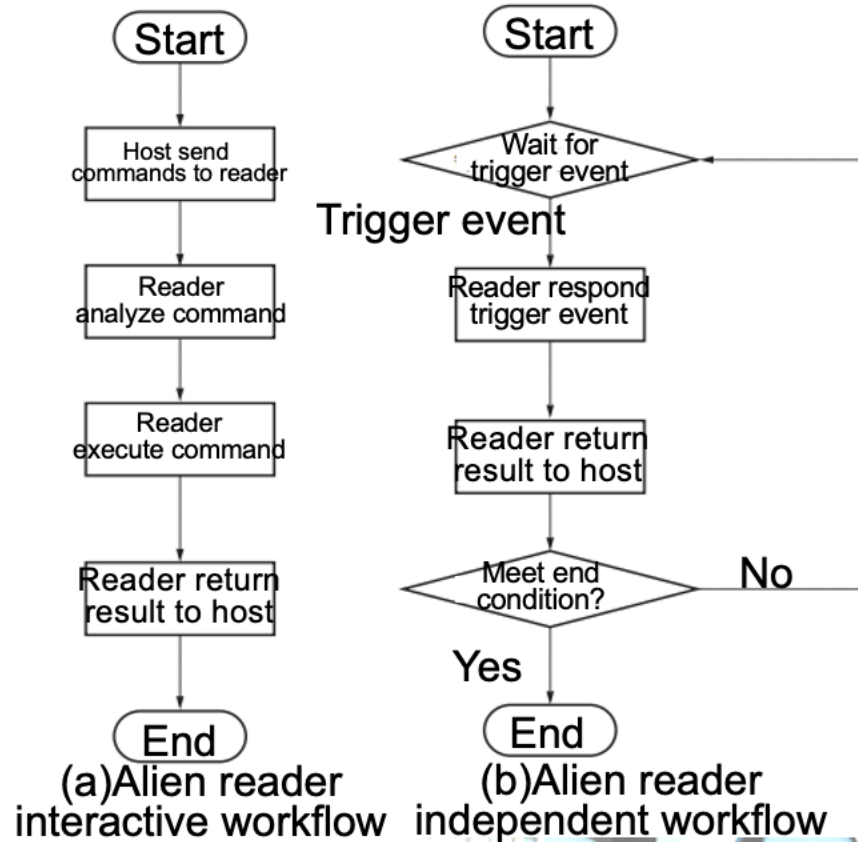
Network service listener class:

NetworkDiscoveryListenerService

Alien RFID Java middleware

Work mode of interactive and independent

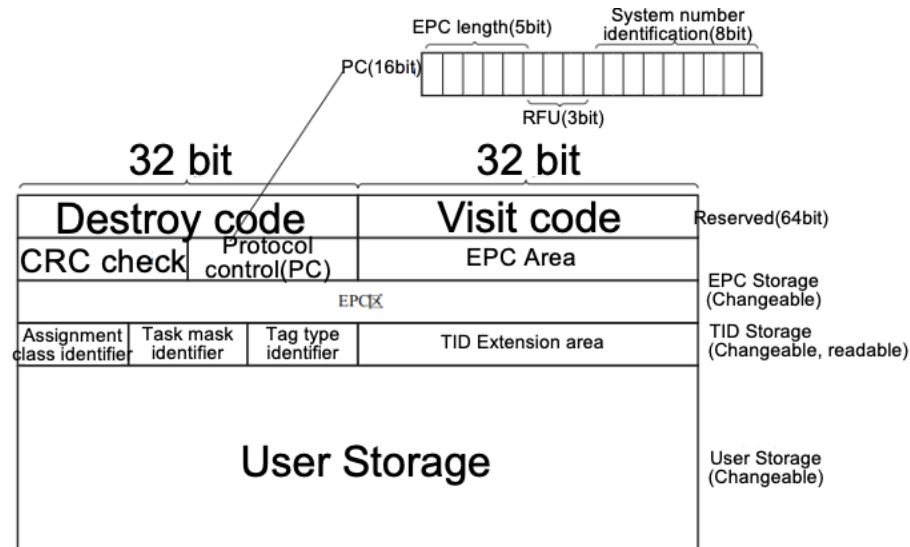
Interactive mode: Readers see each command from host as a single task, execute by order and return result immediately.



Alien RFID Java middleware

Tag Storage Structure

Typical tag storage structure of EPC Class1 Generation2



Tag Mask

Set mask

e.g. only visit tags start with ID="80000040"

Set mask = 32, 0, 80 00 00 40

Reader Operation Full Example

Connect reader by serial port

Through AlienReaderClass1Reader class
e.g. connect through COM1

```
package com.alien.enterpriseRFID.examples;
```

```
import com.alien.enterpriseRFID.reader.*;  
import com.alien.enterpriseRFID.tags.*;
```

```
public class AlienClass1ReaderTest {
```

```
    public AlienClass1ReaderTest() throws AlienReaderException {
```

```
        AlienClass1Reader reader = new AlienClass1Reader();  
        reader.setConnection("COM1");
```

```
        reader.open();
```

//代码解读

Import Required Packages for
Reader and Tag Handling

Create a reader instance and set the
connection port (COM1)

Open the reader connection to
establish communication

Reader Operation Full Example

Connect reader by serial port

```
Tag tagList[] = reader.getTagList();
```

```
if (tagList == null) {  
    System.out.println("No Tags Found");  
}
```

```
else {  
    System.out.println("Tag(s) found:");  
    for (int i = 0; i < tagList.length; i++) {  
        Tag tag = tagList[i];  
        System.out.println("ID:" + tag.getTagID()  
            + ", Discovered:" + tag.getDiscoverTime()  
            + ", Last Seen:" + tag.getRenewTime()  
            + ", Antenna:" + tag.getAntenna()  
            + ", Reads:" + tag.getRenewCount());  
    }  
}
```

```
reader.close();  
}
```

Retrieve the tag list from the reader

Check if any tags were found

If tags exist,, iterate through them and print details such as:

Tag ID

Discovery time (when it was first detected)

Last seen time

Antenna number

Total read count

Closing reader connection

Reader Operation Full Example

Connect reader by serial port

```
public static final void main(String args[]) {  
    try {  
        new AlienClass1ReaderTest();  
    } catch (AlienReaderException e) {  
        System.out.println("Error: " + e.toString());  
    }  
}
```

Try-Catch block for exception handling

Create a new AlienClass1ReaderTest instance, which established a connection to the RFID reader

Handle and display errors if one occurs



Reader Operation Full Example

Reader independent work mode

According to MessageListener messageReceived

```
package com.alien.enterpriseRFID.examples;
```

```
import com.alien.enterpriseRFID.reader.*;  
import com.alien.enterpriseRFID.tags.*;  
import com.alien.enterpriseRFID.notify.*;  
import java.net.InetAddress;
```

```
public class MessageListenerTest implements MessageListener {
```

```
    public MessageListenerTest() throws Exception {
```

```
        MessageListenerService service = new MessageListenerService(4000);  
        service.setMessageListener(this);  
        service.startService();  
        System.out.println("Message Listener has Started");
```

```
        AlienClass1Reader reader = new AlienClass1Reader("COM1");  
        reader.open();  
        System.out.println("Configuring Reader");
```

```
        reader.setNotifyAddress(InetAddress.getLocalHost().getHostAddress(),  
                                service.getListenerPort());  
        reader.setNotifyFormat(AlienClass1Reader.XML_FORMAT);  
        reader.setNotifyTrigger("TrueFalse");  
        reader.setNotifyMode(AlienClass1Reader.ON);
```

Import necessary enterprise RFID libraries to handle the reader communication, tag processing, and event notifications

Define the message listener class

Set up the message listener service on port 4000, reader to process tag read events

Connect to RFID reader: create a new AlienClass1Reader instance and configure it via serial port (COM1).

Configure how the reader reports tag information

The reader sends tag updates every second and exit independent mode after 10 seconds

Reader Operation Full Example

Reader independent work mode

```
reader.autoModeReset();
reader.setAutoStopTimer(1000);
reader.setAutoMode(AlienClass1Reader.ON);

reader.close();
long runTime = 10000;
long startTime = System.currentTimeMillis();
do {
    Thread.sleep(1000);
} while (service.isRunning()
        && (System.currentTimeMillis() - startTime) < runTime);

System.out.println("\nResetting Reader");
reader.open();
reader.autoModeReset();
reader.setNotifyMode(AlienClass1Reader.OFF);
reader.close();
}
```

Enable independent work mode



Reader Operation Full Example

Reader independent work mode

```
public void messageReceived(Message message) {  
    System.out.println("\nMessage Received:");  
    if (message.getTagCount() == 0) {  
        System.out.println("(No Tags)");  
    } else {  
        for (int i = 0; i < message.getTagCount(); i++) {  
            Tag tag = message.getTag(i);  
            System.out.println(tag.toLongString());  
        }  
    }  
}
```

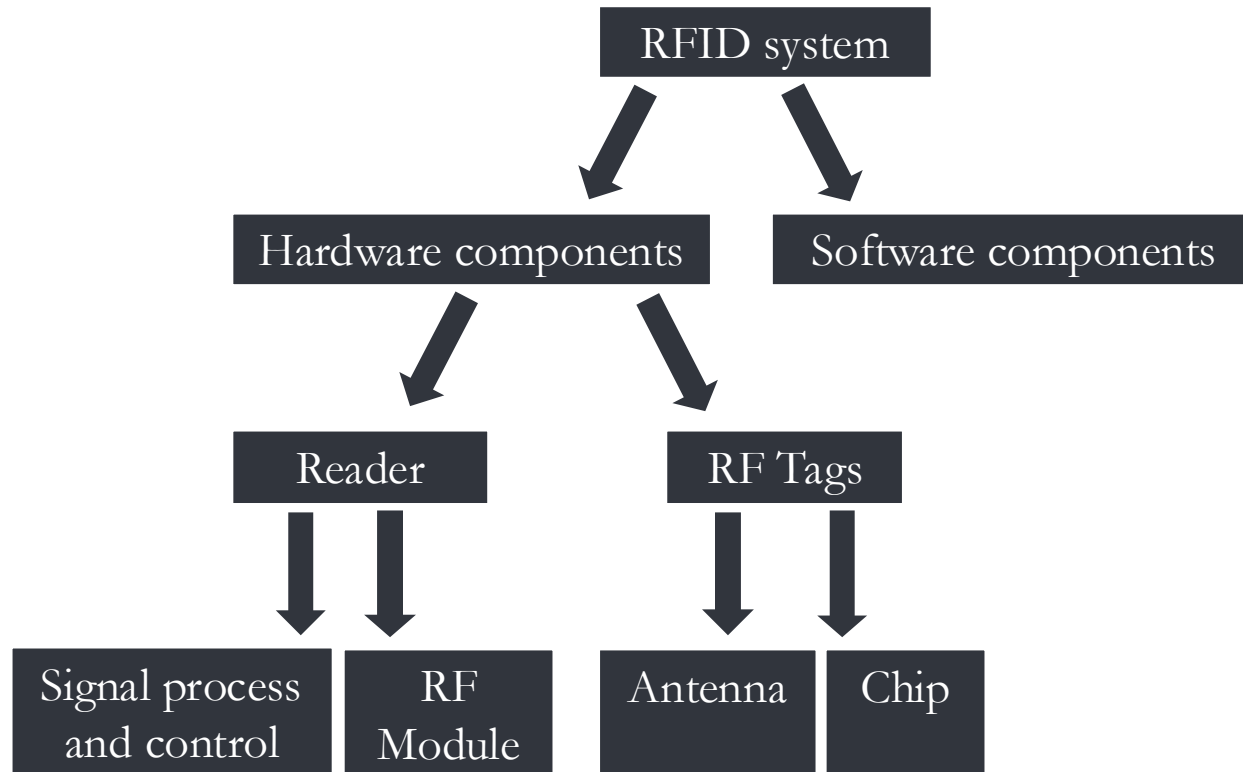
Print tag information

```
public static final void main(String args[]) {  
    try {  
        new MessageListenerTest();  
    } catch (Exception e) {  
        System.out.println("Error:" + e.toString());  
    }  
}
```

Create test class object



RFID System Component Principle Sum Up



Thank You



Dr Fatma Benkhelifa